

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250276261

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Chernov; Gregory Sergeevich

AUTOMATIC SHUTOFF VALVE FOR A WATER FILTER ASSEMBLY

Abstract

A water filter assembly for a domestic appliance includes a filter housing defining a filter chamber, a feed water inlet, and a treated water outlet; a filter element received within the filter chamber defining a feed water chamber and a treated water chamber; a valve plug positioned within the filter housing; and an actuator in operable contact with the valve plug, the actuator being a shape-memory material configured to adjust from a first shape to a second shape in response to a stimulus.

Inventors: Chernov; Gregory Sergeevich (Louisville, KY)
Applicant: Haier US Appliance Solutions, Inc. (Wilmington, DE)
Family ID: 96881730
Appl. No.: 18/591556
Filed: February 29, 2024

Publication Classification

Int. Cl.: B01D35/157 (20060101); C02F1/00 (20230101); F16K31/00 (20060101); F25D23/12 (20060101)

U.S. Cl.:

CPC B01D35/1573 (20130101); C02F1/003 (20130101); F16K31/002 (20130101); F25D23/126 (20130101); B01D2201/16 (20130101); C02F2201/005 (20130101); C02F2307/12 (20130101); F25D2323/121 (20130101)

Background/Summary

FIELD OF THE INVENTION

[0001] The present subject matter relates generally to water filter assemblies, and more particularly to shut off mechanisms within water filter assemblies.

BACKGROUND OF THE INVENTION

[0002] Many known household appliances include water filters or water filter assemblies to remove particulates, dissolved solids, tastes, odors, or other undesirable contaminants of a water supply. Typically, the water filters include consumable filter elements that are replaced at the end of a useful lifetime of the filter. Commonly, the useful life of the filter element is determined by the filter's time in service or the volume of water that has been filtered. Additionally, certain public events such as boil water advisories or other health advisories may render the filter elements compromised or unable to perform required filtering duties.

[0003] However, drawbacks exist in current appliances regarding properly replacing used filters. For instance, many known appliances provide a notification such as a warning light to change the filter after a certain amount of time has passed or a certain volume of water has been cycled through the filter assembly. Moreover, existing methods for restricting flow of water out of the water filter assembly can be ineffective, reversed, or unpredictable, with some requiring extensive equipment leaving less room for filtration material and subsequently reducing the life of the filter.

[0004] Accordingly, a water filter assembly which obviates one or more of the above-mentioned drawbacks would be beneficial. In particular, a water filter assembly with an automatic shutoff valve and actuator would be useful.

BRIEF DESCRIPTION OF THE INVENTION

[0005] Aspects and advantages of the invention will be set forth in part in the following description, or may be obvious from the description, or may be learned through practice of the invention.

[0006] In one exemplary aspect of the present disclosure, a water filter assembly is provided. The water filter assembly may include a filter housing defining a filter chamber, a feed water inlet, and a treated water outlet; a filter element received within the filter chamber defining a feed water chamber and a treated water chamber; a valve plug positioned within the filter housing, the valve plug being movable from an unplugged position to a plugged position; and an actuator assembly in operable contact with the valve plug, wherein the actuator assembly includes a shape-memory material configured to adjust from a first shape to a second shape in response to an electrical stimulus, and wherein the valve plug is in the unplugged position when the shape-memory material is in the first shape and the valve plug moves to the plugged position when the shape-memory material is transformed into the second shape.

[0007] In another exemplary aspect of the present disclosure, a domestic appliance is provided. The domestic appliance may include a cabinet forming a receiving space; a water supply inlet provided on the cabinet and through which water is supplied to the receiving space from an external water supply; and a water filter assembly in fluid communication with the external water supply. The water filter assembly may include a filter housing defining a filter chamber, a feed water inlet, and a treated water outlet; a filter element received within the filter chamber defining a feed water chamber and a treated water chamber; a valve plug positioned within the filter housing; an actuator assembly in operable contact with the valve plug, wherein the actuator assembly includes a shape-memory material configured to adjust from a first shape to a second shape in response to a stimulus; an electric power supply in operable communication with the actuator assembly; and a controller operably connected to the electric power supply, the controller being configured to perform an operation. The operation may include receiving an input signal relating to the water filter assembly; determining that the water filter assembly is compromised after receiving the input signal relating to the water filter assembly; and directing the electric power supply according to a predetermined level to produce the stimulus in response to determining that the water filter

assembly is compromised, wherein the valve plug is moved from an open position to a closed position when the actuator assembly adjusts from the first shape to the second shape in response to the stimulus.

[0008] These and other features, aspects and advantages of the present invention will become better understood with reference to the following description and appended claims. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] A full and enabling disclosure of the present invention, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures.

[0010] FIG. 1 provides a front view of a refrigerator appliance according to an exemplary embodiment of the present disclosure.

[0011] FIG. 2 provides a perspective view of the refrigerator appliance of FIG. 1.

[0012] FIG. 3 provides a front view of the refrigerator appliance of FIG. 1 with doors in an open position.

[0013] FIG. 4 provides a simplified section view of a water filter with an embedded wireless power receiver according to one or more embodiments of the present disclosure which may be incorporated into a refrigerator appliance such as the exemplary refrigerator appliance of FIG. 1.

[0014] FIG. 5 provides a schematic diagram showing electrical communication between the exemplary refrigerator appliance of FIG. 1 and the exemplary water filter of FIG. 4.

[0015] FIG. 6 provides a perspective view of a valve plug and an actuator of the water filter of FIG. 4 according to exemplary embodiments of the present disclosure, with the actuator in a first shape.

[0016] FIG. 7 provides a perspective view of the valve plug and actuator of FIG. 6 according to exemplary embodiments of the present disclosure, with the actuator in a second shape.

[0017] FIG. 8 provides a top view of the valve plug and actuator of FIG. 6 in the first shape.

[0018] FIG. 9 provides a top view of the valve plug and actuator of FIG. 7 in the second shape.

[0019] FIG. 10 provides a side schematic view of a water filter according to exemplary embodiments of the present disclosure, with the actuator in the first shape and the valve plug in a first position.

[0020] FIG. 11 provides a side schematic view of the water filter of FIG. 10 with the actuator in the second shape and the valve plug in a second position.

[0021] FIG. 12 provides a perspective cut away view of a water filter according to exemplary embodiments of the present disclosure including the valve plug and actuator of FIG. 6.

[0022] FIG. 13 provides a close-up perspective view of the bottom of the water filter of FIG. 12 including the valve plug in the first position and the actuator in the first shape.

[0023] FIG. 14 provides a schematic illustration of an exemplary refrigerator appliance in communication with a remote user interface device according to one or more embodiments of the present disclosure.

[0024] FIG. 15 provides a schematic illustration of an exemplary refrigerator appliance in communication with a remote network according to one or more embodiments of the present disclosure.

[0025] FIG. 16 provides a flow chart illustrating a method of operating a refrigerator appliance.

[0026] Repeat use of reference characters in the present specification and drawings is intended to represent the same or analogous features or elements of the present invention.

DETAILED DESCRIPTION

[0027] Reference now will be made in detail to embodiments of the invention, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

[0028] As used herein, the terms “first,” “second,” and “third” may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components. The terms “includes” and “including” are intended to be inclusive in a manner similar to the term “comprising.” Similarly, the term “or” is generally intended to be inclusive (i.e., “A or B” is intended to mean “A or B or both”). In addition, here and throughout the specification and claims, range limitations may be combined and/or interchanged. Such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise. For example, all ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other. The singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.

[0029] Approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “generally,” “about,” “approximately,” and “substantially,” are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value, or the precision of the methods or machines for constructing or manufacturing the components and/or systems. For example, the approximating language may refer to being within a 10 percent margin, i.e., including values within ten percent greater or less than the stated value. In this regard, for example, when used in the context of an angle or direction, such terms include within ten degrees greater or less than the stated angle or direction, e.g., “generally vertical” includes forming an angle of up to ten degrees in any direction, e.g., clockwise or counterclockwise, with the vertical direction V.

[0030] The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” In addition, references to “an embodiment” or “one embodiment” does not necessarily refer to the same embodiment, although it may. Any implementation described herein as “exemplary” or “an embodiment” is not necessarily to be construed as preferred or advantageous over other implementations. Moreover, each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

[0031] FIG. 1 is a front view of an exemplary embodiment of a refrigerator appliance **100**. FIG. 2 is a perspective view of the refrigerator appliance **100**. FIG. 3 is a front view of the refrigerator appliance **100** with fresh food doors **128** thereof in an open position. Refrigerator appliance **100** extends between a top **101** and a bottom **102** along a vertical direction V. Refrigerator appliance **100** also extends between a first side **105** and a second side **106** along a lateral direction L. As shown in FIG. 2, a transverse direction T may additionally be defined perpendicular to the vertical and lateral directions V, L. Refrigerator appliance **100** extends along the transverse direction T between a front portion **108** and a back portion **110**.

[0032] Refrigerator appliance **100** may include a cabinet or housing **120** defining an upper fresh food chamber **122** (FIG. 3) and a lower freezer chamber or frozen food storage chamber **124** arranged below the fresh food chamber **122** along the vertical direction V. An auxiliary food storage chamber may be positioned between the fresh food storage chamber **122** and the frozen food storage chamber **124**, e.g., along the vertical direction V. Because the frozen food storage chamber **124** is positioned below the fresh food storage chamber **122**, refrigerator appliance **100** may be generally referred to as a bottom mount refrigerator. In the exemplary embodiment, housing **120** may also define a mechanical compartment (not shown) for receipt of a sealed cooling system (not shown). Using the teachings disclosed herein, one of ordinary skill in the art will understand that the present technology can be used with other types of refrigerators (e.g., side-by-side) or a freezer appliance as well. Consequently, the description set forth herein is for illustrative purposes only and is not intended to limit the technology in any aspect.

[0033] Refrigerator doors **128** may each be rotatably hinged to an edge of housing **120** for accessing fresh food chamber **122**. It should be noted that while two doors **128** in a “French door” configuration are illustrated, any suitable arrangement of doors utilizing one, two or more doors is within the scope and spirit of the present disclosure. A freezer door **130** may be arranged below refrigerator doors **128** for accessing freezer chamber **124**. In the exemplary embodiment, freezer door **130** is coupled to a freezer drawer (not shown) slidably mounted within freezer chamber **124**. An auxiliary door **127** may be coupled to an auxiliary drawer which may be slidably mounted within the auxiliary chamber.

[0034] Refrigerator appliance **100** may include a valve **160**. Valve **160** may selectively allow a flow of water (e.g., from a municipal water supply) into refrigerator appliance **100**. Thus, valve **160** may be provided on a water supply inlet **162** of refrigerator appliance **100**. Valve **160** may be an electromechanical valve, for instance. However, it should be understood that the valve may be any suitable valve, such as a mechanical valve (with an attached servo motor), a solenoid valve, a motor valve, or the like. Valve **160** may selectively open or close according to a signal input from a controller.

[0035] Operation of the refrigerator appliance **100** can be regulated by a controller **134** that is operatively coupled to a user interface panel **136**. User interface panel **136** may provide selections for user manipulation of the operation of refrigerator appliance **100** to modify environmental conditions therein, such as temperature selections, etc. In some embodiments, user interface panel **136** is proximate a dispenser assembly **132**. In response to user manipulation of the user interface panel **136**, the controller **134** may operate various components of the refrigerator appliance **100**. Operation of the refrigerator appliance **100** may be regulated by the controller **134**, e.g., controller **134** may regulate operation of various components of the refrigerator appliance **100** in response to programming and/or user manipulation of the user interface panel **136**.

[0036] The controller **134** may include a memory and one or more microprocessors, CPUs or the like, such as general or special purpose microprocessors operable to execute programming instructions or micro-control code associated with operation of refrigerator appliance **100**. The memory may represent random access memory such as DRAM, or read only memory such as ROM or FLASH. In one embodiment, the processor executes programming instructions stored in memory. The memory may be a separate component from the processor or may be included onboard within the processor. It should be noted that controllers **134** as disclosed herein are capable of and may be operable to perform any methods and associated method steps as disclosed herein.

[0037] The controller **134** may be positioned in a variety of locations throughout refrigerator appliance **100**. In the illustrated embodiment, the controller **134** is located within the door **128**. In such an embodiment, input/output (“I/O”) signals may be routed between the controller and various operational components of refrigerator appliance **100**. In one embodiment, the user interface panel **136** represents a general purpose I/O (“GPIO”) device or functional block. In one embodiment, the user interface **136** includes input components, such as one or more of a variety of electrical,

mechanical or electro-mechanical input devices including rotary dials, push buttons, and touch pads. The user interface **136** may include a display component, such as a digital or analog display device designed to provide operational feedback to a user. For example, the user interface **136** may include a touchscreen providing both input and display functionality. The user interface **136** may be in communication with the controller via one or more signal lines or shared communication busses. [0038] Using the teachings disclosed herein, one of skill in the art will understand that the present subject matter can be used with other types of refrigerators such as a refrigerator/freezer combination, side-by-side, bottom mount, compact, and any other style or model of refrigerator appliance. Accordingly, other configurations of refrigerator appliance **100** could be provided, it being understood that the configurations shown in the accompanying FIGS. and the description set forth herein are by way of example for illustrative purposes only.

[0039] In some embodiments, e.g., as illustrated in FIG. **4**, the refrigerator appliance **100** includes a water filter (or water filter assembly) **200** connected thereto. For example, the water filter assembly **200** may be coupled to a water supply line (not shown) such that the water filter assembly **200** receives a flow of water **1000**, e.g., which may be referred to as raw or unfiltered water, from the water supply line and provides a flow of water **1002**, e.g., filtered water, to a portion of the water supply line downstream of the filter **200** and/or various fixtures or components of the refrigerator appliance **100** such as an ice maker, water dispenser, etc. The water filter assembly **200** may include a housing or body **202** which defines a liquid receiving space (or filter chamber) **201** with an inlet (e.g., a feed water inlet) **208** opening through the housing **202** into the filter chamber **201** and an outlet (e.g., a treated water outlet) **204** from the liquid receiving space **201** (e.g., from treated water chamber **207**).

[0040] As illustrated in FIG. **4**, unfiltered water **1000** may flow into filter **200** via inlet **204**. The unfiltered water **1000** may flow into a feed water chamber **205**. The water may then pass through a filter medium or element **206**, e.g., a membrane, activated carbon, or other suitable filter medium including combinations of more than one filter media, thereby producing filtered water **1002**. The filtered water **1002** may then be contained within a treated water chamber **207**. The filtered water **1002** may then exit the water filter **200** via outlet **204**. In additional or alternative embodiments, filter medium **206** may include one or more filter electrodes (e.g., an anode, a cathode, etc.) receiving an electrical current to provide a filter action to the water. The one or more electrodes may be positioned within liquid receiving space **201**. For instance, the filter electrodes may perform an electrochemical contaminant reduction within the water to reduce or remove impurities. As will be explained below, electricity for the filter electrodes may be supplied wirelessly (or through mechanical contacts) from refrigerator appliance **100** to water filter **200**.

[0041] Body **202** may define an electronics compartment **203**. In detail, electronics compartment **203** may be fluidly separated from liquid receiving space **201**. As shown schematically in FIG. **4**, electronics compartment **203** may be formed in body **202** adjacent to liquid receiving space **201**. Thus, a portion of body **202** may be positioned between liquid receiving space **201** and electronics compartment **203**. Additionally or alternatively, a top portion, a bottom portion, opposite side portions, and a radially outward portion may be defined by body **202**. Thus, electronics compartment **203** may be said to be positioned within a shell of water filter **200** (e.g., as defined by body **202**).

[0042] One or more electronic components may be housed (e.g., positioned) within electronics compartment **203**. Thus, the one or more electronic components (described below) may be fluidly isolated from liquid receiving space **201**, and thus fluidly isolated from the flow of water received within liquid receiving space **201**. Thus, electronics compartment **203** may define an electronics receiving space. In some instances, electronics compartment **203** may be selectively accessible (e.g., by a user). For instance, a liquid tight access door may be provided on body **202** of water filter **200** providing access to electronics compartment **203**. According to other embodiments, electronics compartment **203** is sealed with respect to each of liquid receiving space **201** and an

ambient atmosphere. As will be described below, water filter **200** may be in wireless communication with refrigerator appliance **100**. However, it should be understood that any suitable form of communication may take place between water filter **200** and appliance **100** (e.g., such as wired communication, contact communication, etc.). Accordingly, power or electricity may be supplied to water filter **200** from appliance **100** in any suitable way, and the disclosure is not limited to the examples provided herein.

[0043] FIG. 5 provides a schematic representation of a water filter (e.g., water filter **200**) and a refrigerator appliance (e.g., refrigerator appliance **100**). In some embodiments, a wireless power transmitter **312** is provided to supply power to one or more electric devices within refrigerator appliance **100**. In particular, wireless power transmitter **312** may be mounted on or within cabinet **120**, in operable communication (e.g., wireless or contact-free communication) with water filter **200** to transmit an electromagnetic field thereto, which may then power one or more electronic components. For instance, wireless power transmitter **312** may transmit an electromagnetic field to an antenna (described below) provided within electronics compartment **203**. Although power (or electricity) transfer between appliance **100** and water filter **200** is described as being wireless herein, it should be understood that some embodiments may include additional or alternative forms of connectivity, power transfer, or electricity exchange, such as wired connections, contact connections, or the like.

[0044] As shown, wireless power transmitter **312** may be physically spaced apart from water filter **200** at discrete portions (e.g., a docking port for water filter **200**) of refrigerator appliance **100**. When assembled, wireless power transmitter **312** may thus be wirelessly coupled to water filter **200** (e.g., without ever coming into direct or conductive electrical contact with water filter). In turn, an air gap may be maintained between the two. For instance, electronics compartment **203** of water filter **200** may be electrically sealed such that no electrical wires or busses pass through body **202**. For example, when water filter **200** is installed within refrigerator appliance **100**, no direct physical contact is made between components within electronics compartment **203** and wireless power transmitter **312**. Advantageously, a potential failure or leak point may be avoided while still permitting power to be supplied to one or more electrical components provided within water filter **200**.

[0045] Additionally or alternatively, wireless power transmitter **312** may operate (e.g., emit electromagnetic fields) according to one of a plurality of protocols. The plurality of protocols may include, for instance, a near-field communication (NFC) protocol, a Qi protocol, a radiofrequency identification (RFID) protocol, a laser protocol, a microwave protocol, or the like. It should be understood that the list of potential protocols is not exhaustive, and any suitable wireless protocol may be used. For instance, among the plurality of protocols, wireless power transmitter **312** may emit power charging signals (e.g., for wireless charging of electronic components), wireless authentication signals (e.g., to confirm that water filter **200** includes the capability to receive wireless charging), data transfer signals (e.g., wireless signals including data related to components of water filter **200**), or the like.

[0046] With reference to FIG. 5, an antenna **210** may be provided within the water filter **200**. In particular, antenna **210** may be embedded in housing **202** of water filter **200**. For example, in some embodiments, antenna **210** is positioned within electronics compartment **203**. Antenna **210** may be configured to receive an electromagnetic field **212** (e.g., emitted by wireless power transmitter **312** within refrigerator appliance **100**). Accordingly, antenna **210** may be configured to receive the electromagnetic field according to an associated protocol (e.g., through which wireless power transmitter **312** is operated), as would be understood.

[0047] The received electromagnetic field **212** may be converted to a usable signal via a power harvesting circuit **214**. In detail, power harvesting circuit **214** may be provided within electronics compartment **203**. Power harvesting circuit **214** may be electronically connected with antenna **210**. Accordingly, electromagnetic field **212** may be transmitted to power harvesting circuit **214** via one

or more electric connections. Power harvesting circuit **214** may then generate an electric current or voltage and transmit the electric current or voltage to an onboard controller. According to some embodiments, the onboard controller is integral with power harvesting circuit **214**. However, a separate, dedicated controller may be included within water filter **200** (e.g., within electronics compartment **203**).

[0048] Generally, wireless power transmitter **312** and antenna **210** may be configured to exchange an electromagnetic field that generates an electrical current. For instance, wireless power transmitter **312** may transmit an electromagnetic field (e.g., as initiated by controller **134**) that is received at antenna **210** (which may also be referred to as a wireless power receiver). At antenna **210**, an electrical current or voltage may be generated and, subsequently, transmitted to the onboard controller (e.g., through an on-board rectifying circuit or activation circuit). For instance, the electromagnetic field may induce an electrical current at power harvesting circuit **214** that can be received by the onboard controller (e.g., within power harvesting circuit). During operations, the onboard controller may be powered by power harvesting circuit **214**.

[0049] As would be understood, the wireless power transmitter **312** and power harvesting circuit **214** may be a matched pair of resonant induction coils. Nonetheless, it is understood that any other suitable wireless power transmission method (e.g., inductive coupling, capacitive coupling, etc.) may be used.

[0050] As mentioned above, water filter **200** may include one or more electronic components **216**. In detail, the one or more electronic components **216** may include a light emitting diode (LED), an electrode (e.g., a filter electrode for performing an ionizing reaction to the water), an electrode life sensor (e.g., to determine a remaining life or efficiency of the electrode), a water volume sensor, a water cycle sensor, or the like. For example, an electrode life sensor may include a voltage sensor on the electrode, a water quality sensor provided adjacent to the electrode, a chemical reaction sensor, or the like. The one or more electronic components may be operably coupled with power harvesting circuit **214** (e.g., via positive and negative voltage connections). Accordingly, the electrical current or voltage generated at power harvesting circuit **214** may be transmitted to the one or more electrical components **216** to provide power thereto. Additionally or alternatively, the one or more electrical components **216** may include a battery (e.g., a rechargeable battery). For instance, power harvesting circuit **214** may have a battery formed integrally therein. Thus, the electrical signal or voltage may be stored within the battery to be delivered to the one or more electrical components **216** at selected times. Additionally or alternatively, the one or more electrical components may include a shutoff valve assembly **220**, described in more detail below.

[0051] Referring now to FIGS. **6** through **13**, shutoff valve assembly **220** will be described in detail. Shutoff valve **220** may define an axial direction A, a radial direction R, and a circumferential direction C. Shutoff valve assembly **220** may be provided within water filter **200**. For instance, shutoff valve assembly **220** may operate to selectively shutoff, close off, cap, or otherwise restrict water flow through water filter **200** (e.g., in response to an input stimulus or signal). Accordingly, in some instances, shutoff valve assembly **220** may be a one-way valve. For instance, shutoff valve assembly **220** may move in one direction, such as from an open position to a closed position, as will be described.

[0052] FIGS. **6** and **8** show shutoff valve assembly **220** in a first position, and FIGS. **7** and **9** show shutoff valve assembly **220** in a second position. Referring first to FIGS. **6** and **8**, shutoff valve assembly **220** may include a valve plug **222**. Valve plug **222** may be positioned within filter body **202**. For instance, valve plug **222** may be positioned within treated water chamber **207**.

Additionally or alternatively, valve plug **222** may be positioned adjacent to treated water outlet **204**. For instance, valve plug **222** may be suspended above treated water outlet **204** (e.g., by an actuator, described below). Thus, while water filter **200** is operational (e.g., functioning or performing filtration and releasing filtered water), valve plug **222** may not restrict the flow of filtered water **1002** out via treated water outlet **204**.

[0053] Valve plug **222** may include a plug base **224**. Plug base **224** may be conical in shape, tapering towards treated water outlet **204**. For instance, a top of plug base **224** may define a first base diameter **B1** and a bottom of plug base **224** may define a second base diameter **B2**. Second base diameter **B2** may be shorter than first base diameter **B1**. Similarly, treated water outlet **204** may have a funnel shape corresponding to the shape of plug base **224**. Accordingly, as will be described further below, when valve plug **222** is released, treated water outlet **204** may be plugged so as to restrict water from flowing out of water filter **200**. Further, it should be noted that plug base **224** as described herein is provided by way of example only, and plug base **224** may have any suitable shape (e.g., cylindrical, squared, tapered, trapezoidal, etc.).

[0054] Valve plug **222** may include a stem **226**. Stem **226** may extend from plug base **224** (e.g., along the axial direction **A**). For instance, stem **226** may extend from the top of plug base **224** along the axial direction **A**. Stem **226** may define a stem diameter **S1**. Stem diameter **S1** may be less than each of first base diameter **B1** and second base diameter **B2**. Stem **226** may be positioned at or near a radial center of plug base **224**. However, stem **226** as described herein is provided by way of example only, and stem **226** may have any suitable shape or location with respect to plug base **224**. For instance, two or more stems **226** may be incorporated in certain embodiments.

[0055] Valve plug **222** may include a plug head **228**. Plug head **228** may be provided at a distal end of stem **226** (e.g., opposite plug base **224**). Thus, stem **226** may connect plug base **224** with plug head **228**. Plug head may define a head diameter **H1**. Head diameter **H1** may be greater than stem diameter **S1**. For instance, plug head **228** may extend (e.g., along the radial direction **R**) away from stem **226**. Plug head **228** may thus be cylindrical in shape. However, plug head **228** as described herein is provided by way of example only, and plug head **228** may have any suitable shape or location with respect to stem **226**.

[0056] Plug head **228** may define an under surface **230**. For instance, stem **226** may extend from under surface **230** of plug head **228**. Under surface **230** may face plug base **224**. Thus, under surface **230** may be provided radially around stem **226**. As will be described below, under surface **230** may selectively rest upon an actuator when in the first position (e.g., as shown in FIG. 6).

[0057] Shutoff valve assembly **220** may include an actuator (or actuator assembly) **232**. Actuator **232** may be in operable contact with valve plug **222**. In detail, actuator **232** may hold valve plug **222** in the first position (FIG. 6). As such, actuator **232** may be positioned around valve plug **222** (e.g., along the circumferential direction **C**). Actuator **232** may be or include a shape-memory material. For instance, a material of actuator **232** may be configured to change shape in response to a stimulus (e.g., an electrical stimulus). Thus, as will be described, actuator **232** may have, define, or otherwise be in a first shape corresponding to the first position of valve plug **222** (FIGS. 6 and 8). Actuator **232** may then selectively change to a second shape corresponding to the second position of valve plug **222** (FIGS. 7 and 9).

[0058] As used herein, the term “shape-memory material” includes various materials including alloys also known in the art as smart metals, memory metals, memory alloys, muscle wires, or smart alloys whose physical shape or length changes with temperature changes. For instance, shape-memory alloys (SMAs) may include a family of metal alloys that possess shape memory and/or pseudoelasticity effects. The shape memory effect may be an ability of a deformed material to return to an original or “trained” shape in response to certain environmental conditions. For at least one example, when a temperature of an exemplary SMA is at an “operating range” (e.g., below an Austenite Start Temperature [**A.sub.s**]), the SMA is soft and pliable and may be easily deformed without an immediate elastic response. Once heated to a temperature above **A.sub.s**, the SMA may initiate a transformation to the trained shape. The transformation may be deemed complete once the temperature is above a transformation temperature (Active Austenite Finish Temperature [**A.sub.f**]). The difference between **A.sub.s** and **A.sub.f** may be between about 15° C. and about 20° C. However, the difference between **A.sub.s** and **A.sub.f** may vary depending on the material composition and heat treatment of the SMA. After the transformation, the shape may

remain in the trained shape unless or until an external load is applied thereto.

[0059] For example, actuator **232** may have a straight, shorter, or more slender length at a first temperature (e.g., an ambient temperature), as shown in FIG. **8**, and a curved, bulbous, or longer length at higher temperatures (e.g., at an actuation temperature such as A.sub.f), as shown in FIG. **9**. The curving or lengthening of actuator **232** at higher temperatures may thus move at least a portion actuator **232** with respect to valve plug **222**. The shape-memory material may be made from various alloy combinations that exhibit the desired changes at the anticipated temperatures. For example, the shape-memory material may include copper-zinc-aluminum-nickel, copper-aluminum-nickel, nickel-titanium, or other alloys of zinc, copper, gold, and iron.

[0060] Actuator **232** may thus be a shape-memory wire. Actuator **232** may include a first leg **234** and a second leg **236**. First leg **234** may be disposed on a first side of valve plug **222** while second leg is disposed on a second side of valve plug (e.g., opposite the first side). In some instances, actuator **232** may be described as surrounding or at least partially encircling valve plug **222**. Moreover, actuator **232** may extend predominantly along the radial direction R and the circumferential direction C (e.g., within treated water chamber **207**). Additionally or alternatively, actuator **232** may be spaced a predetermined distance from treated water outlet **204** (e.g., along the axial direction A). In particular, a distance D1 between actuator **232** and treated water outlet **204** may be greater than a total length L of valve plug **222** (e.g., as shown in FIG. **10**).

[0061] With specific reference to FIGS. **6** and **8**, actuator **232** is in the first shape. As such, first leg **234** and second leg **236** may be positioned relatively close to each other. In detail, a distance W1 between first leg **234** and second leg **236** may be less than head diameter H1 of plug head **228**. As can be seen, each of first leg **234** and second leg **236** may thus be positioned underneath plug head **228** (e.g., along the axial direction A). In other words, first leg **234** and second leg **236** may be in contact with under surface **230** of plug head **228**. Additionally or alternatively, each of first leg **234** and second leg **236** may be provided between plug head **228** and plug base **224**. Actuator **232** may thus retain valve plug **222** in the first position (FIGS. **6** and **10**) when actuator **232** is in the first shape.

[0062] With specific reference to FIGS. **7** and **9**, actuator **232** is in the second shape. As such, first leg **234** and second leg **236** may be positioned relatively far away from each other. In detail, a distance W2 between first leg **234** and second leg **236** may be greater than head diameter H1 of plug head **228**. As can be seen, each of first leg **234** and second leg **236** may be moved out from underneath plug head **228** (e.g., outward along the radial direction R). In other words, plug head **228** may no longer be restrained by actuator **232**. Valve plug **222** may thus move from the first position (FIGS. **6** and **10**) to the second position (FIGS. **7** and **11**). Valve plug **222** may close treated water output **204** and restrict water from flowing out of water filter **200**. In additional or alternative embodiments, actuator **232** may move (e.g., push, pull, twist, or otherwise directly adjust) valve plug **222** when activated, changed, or adjusted from the first position to the second position.

[0063] As described above, actuator **232** may include a shape-memory material, such as a shape-memory alloy. Thus, actuator **232** may be induced to change from the first shape to the second shape according to an input or stimulus. The stimulus may be generated by a heat source. For one example, the heat source is an electrical power supply (e.g., such as power harvesting circuit **214**). Power harvesting circuit **214** may be operably coupled with actuator **232** via one or more electrical connections. Power harvesting circuit **214** may then selectively produce or supply an electrical impulse or input to actuator **232** (e.g., the shape-memory wire). The electrical input may generate a temperature change within actuator **232**. The temperature change in turn may alter actuator **232** from the first shape to the second shape. It should be understood that additional or alternative heat sources may be utilized to provide the stimulus to actuator **232**, and the disclosure is not limited to the examples provided herein.

[0064] In some instances, a protective sheath or covering may be provided around actuator **232**.

For instance, a temperature insulating sleeve (e.g., a polymer sleeve) may be provided around actuator **232**. Accordingly, as actuator heats up, the heat may be restricted from being transferred to the water within treated water chamber **207**. The sleeve may thus prevent localized instances of water boiling within water filter **200**, and may reduce an amount of energy required to reach activation temperature, thus reducing an activation time.

[0065] Shutoff valve assembly **220** may include a heating wire **238**. For instance, the heat source may include heating wire **238** to provide the stimulus (e.g., heat) to actuator **232**. Accordingly, heating wire **238** may be operably connected with actuator **232**. Heating wire **238** may include any suitable conducting material sufficient to provide the heat required to adjust actuator **232** from the first shape to the second shape. For one example, heating wire **238** may be made from a nickel-chromium alloy. Accordingly, if the power or electrical input to water filter **200** is insufficient to generate the required heat within actuator **232**, heating wire **238** may be added to ensure proper execution of actuator **232**.

[0066] Referring briefly to FIGS. **12** and **13**, shutoff valve assembly **220** is shown in an exemplary position within water filter **200**. As seen, shutoff valve assembly **220** may be positioned within treated water chamber **207**. In particular, shutoff valve assembly **220** may be positioned relatively proximate treated water outlet **204**. Accordingly, valve plug **222** may be induced toward treated water outlet **204** according to a number of ways. In some instances, the natural flow of water toward treated water outlet **204** carries valve plug **222** toward treated water outlet **204** (e.g., after actuator **232** is adjusted to the second shape). Additionally or alternatively, valve plug **222** may be urged toward treated water outlet **204** via gravity. In still further embodiments, one or more of a spring, a flap, a diaphragm, a ball valve, or the like may be incorporated to urge, push, direct, or otherwise guide valve plug **222** toward treated water outlet **204** (see FIG. **7**).

[0067] FIG. **14** schematically illustrates the refrigerator appliance **100** communicating with a remote user interface device **1100**. Also shown (but not numbered) in FIG. **14** is a user such as may interact with the remote user interface device **1100**, e.g., via a user interface **1102** of the remote user interface such as a touchscreen in the illustrated embodiment. For example, the remote user interface device **1100** may be a device such as a cell phone, smart phone, smart assistant, or any similar device in operative communication with the controller **134** via a wireless connection. As shown in FIG. **14**, the refrigerator appliance **100**, and in particular, controller **134** thereof, may be configured to communicate with a separate device external to the appliance **100**, such as a communications device or other remote user interface device **1100**. The remote user interface device **1100** may be a laptop computer, smartphone, tablet, personal computer, wearable device, smart home system (such as a smart assistant speaker), or various other suitable devices. The refrigerator appliance **100** may include a network communication module, e.g., a wireless communication module, for communicating with the remote user interface device **1100**. In various embodiments, a network communication module may include a network interface such that the controller **134** of the refrigerator appliance **100** can connect to and communicate over one or more networks with one or more network nodes. A network communication module may also include one or more transmitting, receiving, or transceiving components for transmitting/receiving communications with other devices communicatively coupled with refrigerator appliance **100**. The network communication module may be in communication with, e.g., coupled or connected to, the controller **134** to transmit signals to and receive signals from the controller **134**.

[0068] As schematically illustrated in FIG. **14**, the refrigerator appliance **100** may be configured to communicate with the remote user interface device **1100** either directly or through a network **2000** (e.g., a smart home network). Thus, in various embodiments, the refrigerator appliance **100** and the remote user interface **1100** may be configured to communicate wirelessly with each other or with the network **2000**. The network **2000** may be or include various possible communication connections and interfaces, e.g., such as Zigbee, BLUETOOTH®, WI-FI®, or any other suitable communication connection. The remote user interface device **1100** may include a memory for

storing and retrieving programming instructions. For example, the remote user interface device **1100** may be a smartphone operable to store and run applications, also known as “apps,” and may include a remote user interface provided as a smartphone app. Additionally or alternatively, multiple remote user interface devices **1100** may be connected with refrigerator appliance **100**. For instance, one or more smart phones, one or more smart assistant devices (smart speakers), or the like may be simultaneously connected with refrigerator appliance **100** (e.g., through network **2000**). [0069] FIG. **15** provides a graphical representation of a system **170** including appliance **100** and user interface device **1100**. For instance, the system **170** may include network **2000**. One or more remote sources **300** (e.g., news sources, social media networks, search engines, etc.) may be connected to network **2000**. Thus, information from the one or more remote sources **300** may be selectively provided to appliance **100**, user interface device **1100**, or additional remote devices. The network may include a remote server **2002**. The remote server **2002** may collect, compile, organize, analyze, distribute, etc., the information collected from remote sources **300**. Additionally or alternatively, remote server **2002** may develop or otherwise formulate informational data files **302** (e.g., guides, tips, instructions, etc.). The informational data files may be selectively distributed to appliance **100**, user interface device **1100**, or the like, as will be explained in further detail below. [0070] FIG. **16** depicts a flow diagram of an example method **400** of operating a domestic (e.g., refrigerating) appliance according to a specific embodiment of the present disclosure. The controller **150** of appliance **100** may be configured or programmed to implement method **400**. In addition, FIG. **16** depicts steps performed in a particular order for purposes of illustration and discussion. Those of ordinary skill in the art, using the disclosures provided herein, will understand that the steps of any of the methods disclosed herein may be modified, adapted, expanded, omitted, and/or rearranged in various ways without deviating from the scope of the present disclosure. Additionally or alternatively, a separate dedicated controller may be configured to perform method **400**. Further still, method **400** may be applied to any suitable domestic appliance configured to supply water (e.g., liquid water, ice, etc.) to one or more users (e.g., a stand-alone ice dispenser, a stand-alone beverage dispenser, a water cooler, etc.).

[0071] At step **402**, method **400** may include receiving an input signal relating to the water filter assembly. For instance, the input signal may include information from one or more remote sources relating to a quality of water from a municipal water supply to the water supply inlet of the appliance (or to the water filter assembly). In detail, the appliance (e.g., refrigerator appliance **100**) may, through a wireless connection to a wireless network, receive at least one informational notification regarding a potential or impending water advisory, such as a boil water advisory. The remote sources may include local or regional news channels or outlets, utilities agencies, health boards, social media networks (e.g., Twitter®, Facebook®, etc.), community message boards, or the like. For instance, the controller of the appliance may periodically scan the one or more remote sources for information relating to water quality issues. Additionally or alternatively, the appliance may include one or more algorithms configured to retrieve or import water-quality specific alerts from the one or more sources.

[0072] The one or more remote sources may include a separate remotely connected appliance. For instance, a plurality of appliances may be interconnected via a cloud or wireless network, for example through a remote server. The plurality of appliances may be provided remotely to each other (e.g., in different households or physical locations). Additionally or alternatively, each of the plurality of appliances may belong to or be registered to different unique users. Accordingly, a first remote appliance may receive an input regarding a water quality issue directly from a user. For example, a first user associated with the first remote appliance independently discovers the water quality issue and inputs the discovered information to the first remote appliance. The first remote appliance may then upload the water quality issue to the remotely connected server or cloud. The remotely connected server or cloud may then distribute or communicate the water quality issue to a second remote appliance. Additionally or alternatively, the input signal may include a usable life of

the water filter or water filter assembly. For instance, the input signal may be a signal that the water filter is expired and needs to be changed.

[0073] At step **404**, method **400** may include determining that the water filter assembly is compromised after receiving the input signal relating to the water filter assembly. As mentioned above, the input signal may include one or more of a boil water advisory, a health advisory, a containment issue, an expiration date of the water filter, or the like. Accordingly, method **400** may determine that the filter is no longer effective at producing filtered water up to a required standard.

[0074] At step **406**, method **400** may include directing the heat source to a predetermined level to produce a stimulus. In detail, after determining that the water filter is compromised, method **400** may shut off, close off, or otherwise deactivate the water filter. The heat source (or electrical input) may be directed to produce and provide heat to an actuator (e.g., actuator **232**) to adjust from a first position to a second position (e.g., as described above). Upon adjusting from the first position to the second position, a valve plug (e.g., valve plug **222**) may be released within the water filter to plug an outlet (or an inlet) of the water filter, effectively rendering the water filter unusable.

[0075] In some instances, the stimulus includes the electrical input or impulse to the actuator. The electrical input may include a voltage level and a current level. For instance, some systems may incorporate a fixed voltage level of electrical inputs while varying current levels. Similarly, some systems may incorporate a fixed current level of electrical inputs while varying voltage levels. Method **400** may include determining that at least one of the voltage level or a current level through the actuator has changed by a predetermined amount. For instance, one or more sensors (e.g., resistors) may be included and electrically connected with the actuator to sense, measure, or otherwise determine the voltage or current levels within the actuator. The changing of the voltage or current level may signify that the actuator has successfully adjusted from the first shape to the second shape.

[0076] Upon determining that at least one of the voltage or current levels has changed by the predetermined amount, method **400** may conclude that the actuator has successfully adjusted and the valve plug has been released within the water filter. Accordingly, method **400** may direct the heat source to an off position. For instance, method **400** may stop the electrical input to the actuator or otherwise stop producing heat at the actuator (e.g., via the heat source, the electrical power supply, the heating wire, etc.). Advantageously, power may be saved after immediately determining the completion of the actuator shape change.

[0077] This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

Claims

1. A water filter assembly for a domestic appliance, the water filter assembly defining an axial direction, a radial direction, and a circumferential direction, the water filter assembly comprising: a filter housing defining a filter chamber, a feed water inlet, and a treated water outlet; a filter element received within the filter chamber defining a feed water chamber and a treated water chamber; a valve plug positioned within the filter housing, the valve plug being movable from an unplugged position to a plugged position; and an actuator assembly in operable contact with the valve plug, wherein the actuator assembly comprises a shape-memory material configured to adjust from a first shape to a second shape in response to an electrical stimulus, and wherein the valve

plug is in the unplugged position when the shape-memory material is in the first shape and the valve plug moves to the plugged position when the shape-memory material is transformed into the second shape.

2. The water filter assembly of claim 1, further comprising: an electric power supply in operable communication with the actuator assembly; and a controller operably connected to the electric power supply, the controller being configured to perform an operation, the operation comprising: receiving an input signal relating to the water filter assembly; determining that the water filter assembly is compromised after receiving the input signal relating to the water filter assembly; and directing the electric power supply according to a predetermined level to produce the electrical stimulus in response to determining that the water filter assembly is compromised, wherein the valve plug is moved from the unplugged position to the plugged position when the shape-memory material adjusts from the first shape to the second shape in response to the electrical stimulus.

3. The water filter assembly of claim 2, wherein the operation further comprises: determining that at least one of a voltage or a current through the actuator assembly has changed by a predetermined amount; and directing the electric power supply to an off position.

4. The water filter assembly of claim 1, further comprising: an insulating sleeve positioned around the shape-memory material.

5. The water filter assembly of claim 1, wherein the valve plug comprises: a plug base; a stem extending from the plug base along the axial direction; and a plug head provided at a distal end of the stem, wherein a head diameter of the plug head is greater than a stem diameter of the stem.

6. The water filter assembly of claim 5, wherein the actuator assembly comprises a shape-memory wire comprising a first leg disposed on a first side of the valve plug and a second leg disposed on a second side of the valve plug opposite the first leg.

7. The water filter assembly of claim 6, wherein a distance between the first leg and the second leg of the actuator assembly is less than the head diameter when the actuator assembly is in the first shape, and the distance between the first leg and the second leg of the actuator assembly is greater than the head diameter when the actuator assembly is in the second shape.

8. The water filter assembly of claim 6, wherein each of the first leg and the second leg contacts an under surface of the plug head when the actuator assembly is in the first shape.

9. The water filter assembly of claim 1, wherein the valve plug is suspended within the treated water chamber via the actuator assembly proximate the treated water outlet when the actuator assembly is in the first shape.

10. The water filter assembly of claim 1, further comprising: a heating wire operably connected with the shape-memory material and configured to provide a predetermined amount of heat to the shape-memory material.

11. A domestic appliance comprising: a cabinet forming a receiving space; a water supply inlet provided on the cabinet and through which water is supplied to the receiving space from an external water supply; and a water filter assembly defining an axial direction, a radial direction, and a circumferential direction, the water filter assembly being in fluid communication with the external water supply, the water filter assembly comprising: a filter housing defining a filter chamber, a feed water inlet, and a treated water outlet; a filter element received within the filter chamber defining a feed water chamber and a treated water chamber; a valve plug positioned within the filter housing; an actuator assembly in operable contact with the valve plug, wherein the actuator assembly comprises a shape-memory material configured to adjust from a first shape to a second shape in response to a stimulus; an electric power supply in operable communication with the actuator assembly; and a controller operably connected to the electric power supply, the controller being configured to perform an operation, the operation comprising: receiving an input signal relating to the water filter assembly; determining that the water filter assembly is compromised after receiving the input signal relating to the water filter assembly; and directing the electric power supply according to a predetermined level to produce the stimulus in response to

determining that the water filter assembly is compromised, wherein the valve plug is moved from an open position to a closed position when the actuator assembly adjusts from the first shape to the second shape in response to the stimulus.

12. The domestic appliance of claim 11, wherein the stimulus from the electric power supply is converted to thermal energy within the actuator assembly to increase a temperature of the shape-memory material.

13. The domestic appliance of claim 11, further comprising: an insulating sleeve positioned around the shape-memory material.

14. The domestic appliance of claim 11, wherein the valve plug comprises: a plug base; a stem extending from the plug base along the axial direction; and a plug head provided at a distal end of the stem, wherein a head diameter of the plug head is greater than a stem diameter of the stem.

15. The domestic appliance of claim 14, wherein the actuator assembly comprises a shape-memory wire comprising a first leg disposed on a first side of the valve plug and a second leg disposed on a second side of the valve plug opposite the first leg.

16. The domestic appliance of claim 15, wherein a distance between the first leg and the second leg of the shape-memory wire is less than the head diameter when the actuator assembly is in the first shape, and the distance between the first leg and the second leg of the shape-memory wire is greater than the head diameter when the actuator assembly is in the second shape.

17. The domestic appliance of claim 15, wherein each of the first leg and the second leg contacts an under surface of the plug head when the actuator assembly is in the first shape.

18. The domestic appliance of claim 11, wherein the valve plug is suspended within the treated water chamber via the actuator assembly proximate the treated water outlet when the actuator assembly is in the first shape.

19. The domestic appliance of claim 11, wherein the operation further comprises: determining that at least one of a voltage or a current through the actuator assembly has changed by a predetermined amount; and directing the electric power supply to an off position.

20. The domestic appliance of claim 11, wherein the water filter assembly further comprises: a heating wire operably connected with the shape-memory material and configured to provide a predetermined amount of heat to the shape-memory material.
